LonghornSilicon ACU × KV Cache Engine Integration Audit

LonghornSilicon

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Abstract

We audit the integrated dataflow between the Attention Compute Unit (ACU, block 1) and the KV Cache Engine (KVCE, block 2) of the LonghornSilicon LLM inference accelerator. The audit produced a reference-model bug-fix (`kv-cache-engine@9b1163a`, closes issue #1) that lifted decompressed-K cosine similarity from 0.475 to 0.978 and reduced integrated relative MSE from 17.0 to 0.36 against a dense FP16 baseline. The chip-level integration is unblocked. However, three quality gaps remain in the post-fix state and are tracked here for traceability: a fixed-point boundary clipping at the KVCE input port, a calibration mismatch at network layer 0, and the absence of an end-to-end accuracy measurement on real model outputs. Each is filed as a separate issue against the KVCE repository.

1. Scope and methodology

The audit compares five attention-output paths, computed per 64×64 attention tile on real Qwen2-0.5B traces (seq_len = 512, layers $\{0, 4, 8, 12, 16, 20, 23\}$, 2744 off-diagonal tiles total):

Path	**K source**	**V source**	**S·V matmul precision**
A — *REF*	true (FP16)	true (FP16)	FP16 dense
B — PC alone (no KVCE)	true (FP16)	true (FP16)	PC-routed
C — KVCE alone, FP16 SV	$\hat{K}$ (lossy)	$\hat{V}$ (lossy)	FP16
D — KVCE alone, INT8 SV	$\hat{K}$ (lossy)	$\hat{V}$ (lossy)	INT8
E — Integrated	$\hat{K}$ (lossy)	$\hat{V}$ (lossy)	PC-routed on lossy S

Relative MSE of paths B, C, D, E against the REF baseline isolates the contribution of each block: B measures the ACU in isolation; C and D measure the KVCE in isolation under the two possible routing choices; E measures the integrated chip. Per-tile records carry the precision controller's INT8/FP16 decision on both clean and lossy S, so we can also measure *decision stability* under upstream noise.

The harness lives at `adaptive-precision-attention/analysis/integration_test_kv_pc.py` and runs in under two minutes per layer on an RTX A4000. No tile, attention matrix, or per-head activation is persisted to disk; per-tile scalars accumulate in RAM and dump as a single JSON summary.

2. Initial finding (pre-fix)

The initial smoke pass (layer 0, 392 tiles) returned median relative MSE for the four non-REF paths:

Path	Median rMSE
B — PC alone	0.0001
C — KVCE + FP16 SV	17.01
D — KVCE + INT8 SV	17.05
E — Integrated	17.05

The ACU's output error in isolation matched the original precision controller paper's 91–97% INT8-safe claim to four decimal places. The KVCE-in-the-loop output error was 170,000× larger, swamping the FP16-vs-INT8 distinction the precision controller exists to make. Two isolation probes confirmed the problem was upstream of the ACU entirely:

- On the KVCE repository's own `_random_vector` test inputs (which the repo's 184 passing tests use), the median relative MSE of compress–decompress round-trip was 62× for keys and 1.9× for values.
- On charitable $\mathcal{N}(0, 1/\sqrt{d})$ Gaussian inputs (exactly the distribution the Lloyd-Max centroids were designed for), key reconstruction cosine similarity was 0.475 — a decompressed key vector pointed in essentially a random direction relative to the original.

The repository's existing round-trip test used an absolute-integer-space MSE threshold of $\max_{\mathrm{val}}^2 \approx 10^9$ as a pass gate. This is a *doesn't-crash* threshold: literally random output would pass it. There was no test asserting that reconstruction approximated the input.

3. Root cause and remediation

Two bugs in the KVCE reference model, both in the fixed-point arithmetic of the compress/decompress pipeline:

1. **Normalize shift bug.** In `kv_cache_engine_ref.py`, the `normalize()` function divided the coordinate by the stored norm using a left-shift of `coord_frac` (= 12) bits before the integer division. The correct shift is `norm_frac` (= 8) since the norm is stored in Q8.8 format. The 4-bit overshift caused a 16× overscale on all post-normalize coordinates, which then passed through the Walsh–Hadamard rotation and Lloyd–Max quantizer in a regime far outside the centroid design range. After fix, key cosine similarity rose from 0.475 to 0.978.
2. **C++ floor-division parity.** The C++ port used the language's default `/` operator on negative numerators, which truncates toward zero. The Python reference uses `//`, which floors toward negative infinity. The two differ by one on negative inputs and the per-coordinate error accumulated across the 64-element vector. A `floor_div()` helper was added to the C++ port to match the Python semantics bit-exactly.

The fix is a three-line change in the Python reference and an eight-line addition in the C++ port. The fix commit is `kv-cache-engine@9b1163a` (closes issue #1). The repository now ships a quality gate at `sw/reference_model/test_reconstruction_quality.py` that asserts cosine similarity >

0.9 and relative MSE < 0.1 on both K and V paths over Gaussian and `_random_vector` input distributions.

4. Post-fix results

4.1 Aggregate

Re-running the five-path integration harness on 2744 off-diagonal tiles across 7 layers ($\{0, 4, 8, 12, 16, 20, 23\}$) yields:

Path	Median rMSE	p90	p99
B — PC alone (no KVCE)	0.0002	0.0014	0.0250
C — KV alone, FP16 SV	0.3637	3.5709	13.538
D — KV alone, INT8 SV	0.3631	3.5662	13.447
E — Integrated (PC + KV)	0.3631	3.5662	13.447

The integrated median rMSE dropped from 17.0 (pre-fix) to 0.36 (post-fix), a 47× improvement. The KVCE noise floor is now $\sim$ 1800× the precision controller's INT8 routing error, down from $\sim$ 170,000× pre-fix.

4.2 Decision stability

The precision controller's INT8/FP16 decision is computed on $S = Q\hat{K}^\top/\sqrt{d}$ in the integrated path. Compared with the decision computed on clean $S = QK^\top/\sqrt{d}$:

Metric	Value
FP16% on clean S	0.00%
FP16% on lossy S	0.18%
Decisions agree	99.82%
Flipped clean-FP16 → lossy-INT8	0
Flipped clean-INT8 → lossy-FP16	5

The decision function is robust to KVCE-induced perturbation of the score matrix: 99.82% of routing decisions are preserved, and the five flipped tiles flipped *conservatively* (toward FP16, the safe direction).

4.3 Per-layer breakdown

Layer	n	Clean FP16%	Lossy FP16%	rMSE B	rMSE C	rMSE D	rMSE E
0	392	0%	0%	0.0001	2.567	2.576	2.576
4	392	0%	0%	0.0002	0.534	0.534	0.534
8	392	0%	0%	0.0002	0.254	0.254	0.254
12	392	0%	0%	0.0003	0.159	0.160	0.160
16	392	0%	1.28%	0.0003	0.335	0.339	0.339
20	392	0%	0%	0.0004	0.297	0.304	0.304
23	392	0%	0%	0.0004	0.342	0.344	0.344

The 16× spread between layer 0 (2.567) and layer 12 (0.159) is the largest single per-layer variation and is the subject of section 5 item *(b)*.

5. Residual quality gaps

The post-fix integrated rMSE of 0.36 is 47× better than pre-fix but is still $\sim$ 3,600× worse than the ACU in isolation. Three distinct mechanisms contribute, ranked by estimated magnitude:

(a) Q4.12 input clipping at the KVCE boundary

The KVCE input port is specified as int16 Q4.12 fixed-point with representable range ±8.0 at resolution 2^{-12}. Real Qwen2-0.5B post-`k_proj` key activations have absolute maximum approximately ±152 at the layers we measured. The integration harness converts FP16 → Q4.12 at the boundary, which clips approximately 95% of K's magnitude information before any compression is attempted. Per-layer rMSE correlates with K magnitude, consistent with this being the largest single contributor to the residual error. **This is not a KVCE-internal defect**; it is a cross-block specification gap. No document or issue currently records this constraint. Filed as KVCE issue #2; see section 6.

(b) Layer-0 centroid mismatch

The Lloyd–Max centroid set was derived under the assumption that the Walsh–Hadamard-rotated normalized vector has coordinates drawn from $\mathcal{N}(0, 1/\sqrt{d})$. This assumption is well approximated for mid-network activations (layers ≥ 4 in our trace) but poorly approximated for layer 0, whose K/V outputs have not yet been smoothed by the residual stream's accumulating LayerNorms. The result is a 16× rMSE spread across layers, with layer 0 alone contributing the bulk of the corpus median. Candidate remediations include per-layer centroid tables and an adaptive `turbo4-adaptive` mode in the KVCE compression algorithm. Filed as KVCE issue #3.

(c) TurboQuant+ `turbo4` compression noise floor

The current KVCE configuration uses 3-bit PolarQuant plus 1-bit QJL on keys (4.25 bpv) and 3-bit PolarQuant only on values (3.0 bpv). At this compression ratio, per-vector reconstruction error is approximately 0.04 rMSE per coordinate. This error compounds across the 64 V-vectors of the $S \cdot V$ matmul, producing the algorithmic floor that limits how good post-fix integrated rMSE can be. Higher-precision modes (6 bpv `turbo8`, 12 bpv `turbo16`) would lower this floor at the cost of compression ratio. This is a *design property*, not a bug.

6. Known open items

The following items are not blockers for cross-block dataflow but are tracked for traceability through tape-out:

Item	**Severity**	**Issue**	**Doc**
Q4.12 input clipping vs. raw activation magnitudes	high	kv-cache-engine#2	§5(a)
Layer-0 centroid fit (per-layer or adaptive centroids)	medium	kv-cache-engine#3	§5(b)
End-to-end accuracy on real LLM not yet measured	medium	kv-cache-engine#4	§5, §8.3
GQA convention in KVCE ISA	medium	— (architectural)	companion conflicts doc
`SRAM_DEPTH` default is below hot-tier need	low	— (configuration)	companion conflicts doc
`turbo4` noise floor dominates PC FP16 distinction	informational	— (design property)	§5(c)

The first three items are tracked as KVCE GitHub issues with this document referenced as the source. Items four and five are covered in the companion document *Cross-Block Architectural Conflict Register* (`kvce_acu_architectural_conflicts.tex`).

7. Interpretation

Improvement from fix

Metric	Pre-fix	Post-fix	Improvement
K cosine similarity	0.475	0.978	random → accurate
Integrated median rMSE	17.0	0.36	47×
KVCE rMSE / PC rMSE	170,000×	1,800×	94×

Answers to the original questions

Q1. Decision stability under lossy KV. Excellent. 99.82% of precision-controller decisions are unchanged when the input switches from true K to lossy $\hat{K}$. The five tiles that flipped all moved in the conservative direction (toward FP16). The PC decision function is empirically robust to 3-bit PolarQuant plus 1-bit QJL noise.

Q2. Does FP16 routing still help under lossy V? Not materially on the current `turbo4` configuration. The KV quantization rMSE (0.36) is roughly 1,800× the precision controller's routing-induced rMSE (0.0002), so the FP16 vs. INT8 distinction (0.0006 rMSE between paths C and D) is below the KV noise floor. The FP16 path is not actively harmful — the gate fires correctly — but it cannot recover accuracy that the upstream compression has already discarded. The ACU's 1.7×

bandwidth-and-compute claim is intact for *ACU-isolated* workloads (e.g. external K/V supplied at full precision); on the integrated chip, the claim degrades to "bandwidth-and-compute savings with measurable accuracy cost relative to dense FP16 attention."

Q3. Integrated MSE vs. ACU-alone MSE. Path E (integrated) tracks path D (KV+INT8) almost exactly, because the PC routes virtually all tiles INT8 under lossy conditions. The PC contributes neither benefit nor harm in this configuration; it would re-acquire value at higher KVCE precision modes where the K/V noise floor drops below the FP16 vs. INT8 distinction.

Layer-level observations

- Layer 0 is the worst layer (rMSE 2.57): early-layer activations have not been LayerNorm-smoothed and do not match the Lloyd–Max centroid design point.
- Layer 12 is the best (rMSE 0.16): mid-network activations closest to the centroid distribution.
- The 16× spread across layers suggests per-layer or adaptive centroid tables could substantially improve quality at the same compression ratio.

8. Implications per repository

8.1 `adaptive-precision-attention` (this repository)

- The precision controller block is validated: rMSE 0.0002 in isolation; decision stability 99.82% under lossy inputs. No code changes are required.
- The FP16 routing path is not decorative in general — it remains valuable when V is not lossy-quantized (e.g. if a future KVCE variant compresses K only, or under higher-precision modes). On the current `turbo4` configuration, the KV noise dominates and the PC FP16 path provides $<$ 0.2% relative rMSE improvement over INT8.
- Future work: re-run this audit with higher-precision KVCE modes (`turbo8`, `turbo16`) once they are implemented, to test whether the FP16 distinction re-emerges as the KV noise floor drops.

8.2 `LonghornSilicon/kv-cache-engine`

- Fix shipped: `kv-cache-engine@9b1163a` (closes #1).
- Quality gate added: `sw/reference_model/test_reconstruction_quality.py` with cosine > 0.9 and rMSE < 0.1 thresholds. The previous $\max^2_{\text{val}}$ threshold did not bound reconstruction quality.
- Three follow-up issues filed (Q4.12 input range, Layer-0 centroid fit, end-to-end accuracy on real LLM) for the residual quality gaps; see §5 and §6.

8.3 LonghornSilicon chip-level

- ACU integration is unblocked at the dataflow level: the two blocks compose end-to-end with meaningful (non-noise) signal.
- Chip-level accuracy claims (perplexity, downstream task metrics on a real model) have not yet been measured against the integrated dataflow. The per-tile rMSE of 0.36 is suggestive but does not directly bound model-output quality; this measurement is tracked as kv-cache-engine#4.

- The binding constraint for end-to-end attention accuracy on the current `turbo4` configuration is the KV cache compression quality, not the SV matmul precision routing.

9. Reproducing

Verify the reference-model fix in isolation:

```
cd /home/<user>/kv-cache-engine
python3 sw/reference_model/test_reconstruction_quality.py
# Expect: K cosine ~ 0.978, V cosine ~ 0.985 on Gaussian inputs.
```

Re-run the integrated five-path harness:

```
cd /home/<user>/adaptive-precision-attention
python3 analysis/integration_test_kv_pc.py \
        --seq-len 512 --layers 0,4,8,12,16,20,23 \
        --prompts 1 --delete-cache-after
# Expect: integrated median rMSE ~ 0.36, layer-0 outlier ~ 2.57.
# Wall clock: ~2 min per layer on RTX A4000.
```

10. Artifacts on this branch

- `analysis/integration_test_kv_pc.py` — five-path streaming harness, zero-tile-persistence, GQA-aware.
- `analysis/integration_test_kv_pc_stats.json` — smoke-pass results for layer 0.
- `docs/findings/kvce-acu-integration-audit.md` — prior markdown version of this audit, superseded by this LaTeX document.
- `docs/findings/kvce_acu_integration_audit.tex` — this document.
- `docs/findings/kvce_acu_architectural_conflicts.tex` — companion document, cross-block conflict register covering items beyond the immediate integration audit (GQA, SRAM depth, bridge specification, etc.).

Change log

- `kvce-acu-audit-0.1` (2026-05-19): initial markdown audit; KVCE bug surfaced.
- `kvce-acu-audit-0.2` (2026-05-22): post-fix re-run (2744 tiles, 7 layers); promoted to LaTeX; residual quality gaps catalogued; open items filed as KVCE issues.